

The Upper Limb

Directions: Each of the incomplete statements or questions below is followed by four or five suggested completions or answers. Select the one which is BEST in each case.

1. Which of the following bones develops by intramembranous ossification?
 - A. Scapula
 - B. Clavicle
 - C. Humerus
 - D. Ulna
 - E. None of the above
2. Rhomboid fossa is present on
 - A. the inferior surface of the clavicle
 - B. the root of the spine of the scapula
 - C. the anterior aspect of the lower end of humerus
 - D. the posterior aspect of the lower end of humerus
3. In relation to scapula
 - A. the spine is subcutaneous
 - B. inferior angle lies at the level of sixth thoracic vertebra
 - C. lateral border is the thinnest border
 - D. scapular notch transmits suprascapular vessels
 - E. coracoid process can be felt in supraclavicular fossa
4. The humerus presents
 - A. medial epicondyle which is smaller than the lateral epicondyle
 - B. coronoid fossa on the posterior aspect of its lower end
 - C. trochlea with a larger lateral flange
 - D. intertubercular sulcus on the medial side
 - E. anatomical neck which is completely intracapsular

Ans. 1. B 2. A 3. A 4. E

5. In relation to ulna

- A. it is shorter than the radius
- B. it is the main bone which transmits forces from the hand
- C. its head articulates with the capitulum of humerus
- D. posterior border is subcutaneous
- E. gives insertion to pronator quadratus

6. In relation to radius

- A. head articulates with the lateral epicondyle of humerus and the radial notch of ulna
- B. its styloid process lies at a higher level than the styloid process of ulna
- C. shaft is convex laterally
- D. does not give attachment to any of the superficial flexor muscles of forearm
- E. has none of the above properties

7. The clavicle

- A. is seldom fractured
- B. is the first bone to ossify in the body
- C. contains a large medullary cavity
- D. laterally articulates with the coracoid process
- E. has conoid tubercle on the sternal end

8. The following carpal bones lie in the proximal row *except*

- A. scaphoid
- B. lunate
- C. capitate
- D. triquetral
- E. pisiform

9. The first metacarpal bone

- A. articulates proximally with trapezoid
- B. has a secondary ossification centre for its base
- C. lies in the plane of other metacarpal bones
- D. gives attachment to abductor pollicis longus and brevis
- E. forms a saddle joint distally with proximal phalanx

10. Which metacarpal articulates with three carpal bones?

- A. I
- B. II
- C. III
- D. IV
- E. V

Ans. 5. D 6. C 7. B 8. C 9. B 10. B

11. **The infraglenoid tubercle**
- A. lies about 3 cm. below the glenoid cavity
 - B. gives attachment to the long head of biceps brachii
 - C. gives attachment to the lateral head of triceps brachii
 - D. is intracapsular
 - E. has none of the above properties
12. **Which of the following bones has two primary centres of ossification?** *تعميم*
- A. Rib
 - B. Clavicle
 - C. Fibula
 - D. Femur
13. **Which of the following structures passes through the suprascapular notch?**
- A. Subscapular artery
 - B. Circumflex scapular artery
 - C. Suprascapular nerve
 - D. Dorsal scapular nerve
14. **The mammary gland**
- A. consists of 5-10 lobes
 - B. lies entirely deep to deep fascia
 - C. has an axillary tail
 - D. ligaments of Cooper connect it to pectoral fascia.
15. **All the following statements are true for axillary tail of mammary gland except**
- A. lies in superficial fascia
 - B. lies close to axillary vessels
 - C. pierces the deep fascia
 - D. it arises from supero-lateral quadrant of mammary gland.
16. **All the following arteries supply the mammary gland except**
- A. anterior intercostal arteries
 - B. lateral thoracic artery
 - C. superior epigastric artery
 - D. posterior intercostal arteries

Ans. 11. E 12. B 13. C 14. C 15. A 16. C

17. The posterior wall of axilla consists of all the following muscles except

- A. subscapularis
- B. subclavius
- C. latissimus dorsi
- D. teres major

18. The nerve that runs over the medial wall of axilla is

- A. medial pectoral
- B. long thoracic
- C. thoracodorsal
- D. ulnar

19. The axillary lymph nodes drain all the following except

- A. upper limb
- B. mammary gland
- C. body wall above the level of umbilicus
- D. side of neck

20. The musculotendinous cuff is formed by all the following muscles except

- A. supraspinatus
- B. teres minor
- C. long head of triceps
- D. subscapularis
- E. infraspinatus

21. Which of the following muscles has a double nerve supply?

- A. Biceps brachii
- B. Flexor digitorum superficialis
- C. Flexor digitorum profundus
- D. Coracobrachialis
- E. None of the above

22. Which of the following muscles is a flexor and medial rotator of arm?

- A. Teres major
- B. Pectoralis major
- C. Latissimus dorsi
- D. Coracobrachialis
- E. Subscapularis

Chapter 2

23. Which of the following is not a muscle in overhead arm?

- A. Trapezius
- B. Trapezius
- C. Serratus anterior
- D. Deltoid anterior

24. Which of the following is not a rotator of arm?

- A. Teres major
- B. Latissimus dorsi
- C. Posterior deltoid
- D. Anterior deltoid
- E. Sternocleidomastoid

25. Abduction of arm at shoulder joint involves

- A. glenohumeral joint
- B. sternoclavicular joint
- C. acromioclavicular joint
- D. rotator cuff
- E. all the above

26. Which of the following is not a muscle of triceps?

- A. Scapular
- B. Humeral
- C. Clavicular
- D. Ulnar

27. Which of the following is not a muscle of axilla?

- A. Serratus anterior
- B. Pectoralis major
- C. Subscapularis
- D. Subclavius
- E. Deltoid

28. Quadriceps muscle is not a muscle of arm except

- A. Iliotibial band
- B. Tensor fasciae latae

Ans. 23.

Ans. 17. B 18. B 19. D 20. C 21. C 22. B

23. Which of the following pairs of muscles is the most effective in overhead abduction?
- A. Trapezius and serratus anterior
 - B. Trapezius and deltoid
 - C. Serratus anterior and deltoid
 - D. Deltoid and supraspinatus
24. Which of the following muscles is an extensor and lateral rotator of arm?
- A. Teres major
 - B. Latissimus dorsi
 - C. Posterior fibres of deltoid
 - D. Anterior fibres of deltoid
 - E. Sternocostal fibres of pectoralis major
25. Abduction on shoulder joint involves
- A. glenohumeral joint
 - B. sternoclavicular joint
 - C. acromioclavicular joint
 - D. rotation of scapula
 - E. all the above
26. Which of the following bones does *not* give attachment to triceps?
- A. Scapula
 - B. Humerus
 - C. Clavicle
 - D. Ulna
27. Which of the following muscles does not form any boundary of axilla?
- A. Serratus anterior
 - B. Pectoralis major
 - C. Subclavius
 - D. Subscapularis
 - E. Deltoid
28. Quarangular space is bounded by all the following structures *except* the
- A. long head of triceps
 - B. teres minor

Ans. 23. A 24. C 25. E 26. C 27. E

- C. subscapularis
D. infraspinatus
E. surgical neck of humerus
29. Which of the following muscles is *not* a flexor on elbow joint?
A. Biceps brachii
B. Coracobrachialis
C. Brachialis
D. Brachioradialis
E. Pronator teres
30. Which of the following muscles can extend the forearm?
A. Extensor carpi radialis longus
B. Extensor carpi radialis brevis
C. Brachioradialis
D. Anconeus
E. All the above
31. Which of the following muscles is a protractor of scapula?
A. Trapezius
B. Rhomboideus major
C. Pectoralis major
D. Pectoralis minor
E. Deltoid
32. Which of the following muscles does *not* belong to the superficial flexor group of forearm?
A. Pronator teres
B. Flexor carpi radialis
C. Flexor pollicis longus
D. Flexor digitorum superficialis
E. Flexor carpi ulnaris
33. Which of the following structure does *not* pass superficial to flexor retinaculum?
A. Ulnar nerve
B. Ulnar artery
C. Palmaris longus
D. Radial artery
E. Palmar cutaneous branch of median nerve

Ans. 28. D 29. B 30. D 31. D 32. C 33. D

34. True statement about elbow joint is that
A. capsule is thick and strong anteriorly and posteriorly
B. radial collateral ligament is attached to the head of radius
C. synovial membrane is pulled up by articularis cubiti during extension
D. it is a complex joint
E. it has none of the above properties
35. Which of the following is *not* a content of cubital fossa?
A. Tendon of Biceps brachii
B. Median nerve
C. Radial artery
D. Ulnar artery
E. Ulnar nerve
36. Which of the following structures does *not* pass deep to extensor retinaculum?
A. Cephalic vein
B. Posterior interosseous nerve
C. Anterior interosseous artery
D. Extensor digiti minimi
E. Extensor carpi ulnaris
37. Which of the following combination of muscles is best for flexion of wrist?
A. Flexor carpi ulnaris and flexor carpi radialis
B. Flexor carpi ulnaris and flexor digitorum profundus
C. Flexor carpi radialis and flexor digitorum superficialis
D. Flexor digitorum profundus and flexor pollicis longus
38. Which of the following is a prime flexor of metacarpophalangeal joint?
A. Flexor digitorum superficialis
B. Flexor digitorum profundus
C. Flexor carpi radialis
D. Lumbricals
E. None of the above
39. Which of the following is a prime flexor of distal interphalangeal joint?
A. Flexor digitorum superficialis

Ans. 34. E 35. E 36. A 37. A 38. D

- B. Flexor digitorum profundus
C. Lumbricals
D. Palmar interossei
E. None of the above
40. **The fifth metacarpophalangeal joint**
A. is a hinge joint
B. has a thick dorsal ligament
C. is abducted by fourth dorsal interosseous
D. is extended by extensor digitorum
41. **Which of the following is a prime extensor of metacarpophalangeal joint of thumb?**
A. Extensor pollicis longus
B. Extensor pollicis brevis
C. Abductor pollicis longus
D. First lumbrical
E. First dorsal interosseous
42. **The second lumbrical muscle is supplied by**
A. digital branch of median nerve
B. superficial branch of ulnar nerve
C. deep branch of ulnar nerve
D. both median and ulnar nerve
43. **All the muscles on the back of free upper limb are supplied by**
A. median nerve
B. ulnar nerve
C. radial nerve
D. radial and ulnar nerves
E. radial and median nerves
44. **Abduction of thumb occurs on**
A. proximal interphalangeal joint
B. metacarpophalangeal joint
C. carpometacarpal joint
D. intercarpal joint
45. **Adductor pollicis is supplied by**
A. median nerve

Ans. 39. B 40. D 41. B 42. A 43. C 44. C

B. d
C. s
D. r

46. **Whi
ring**

A.
B.
C.
D.

47. **Hyl**

A.
B.
C.
D.
E.

48. **Th**

A.
B.
C.
D.

49. **T**

A
B
C
D

50. **T**

A
B
C
D

51.

Ans.

- B. deep branch of ulnar nerve
C. superficial branch of ulnar nerve
D. radial nerve
46. Which of the following statements is *not* true for adduction of ring finger?
A. Ring finger moves laterally in adduction
B. It involves movement on metacarpophalangeal joint
C. It is done by palmar interosseous muscle
D. Median nerve is responsible for it
47. Hypothenar muscles are supplied by
A. ulnar nerve only
B. median nerve only
C. median and ulnar nerves
D. radial nerve only
E. radial and ulnar nerves
48. The medial pectoral nerve
A. arises from medial cord
B. pierces the clavipectoral fascia
C. topographically situated medial to lateral pectoral nerve
D. supplies pectoralis minor muscle only
49. The thoracodorsal nerve
A. is a branch of lateral cord
B. is accompanied by a branch of posterior circumflex artery
C. supplies latissimus dorsi muscle
D. crosses superficial to axillary vein
50. The axillary nerve supplies all the following structures *except*
A. deltoid
B. teres major
C. shoulder joint
D. skin over lower half of deltoid
51. The long thoracic nerve supplies
A. pectoralis minor
B. teres major
C. subscapularis
D. serratus anterior

Ans. 45. B 46. D 47. A 48. A 49. C 50. B 51. D

52. The root value of ulnar nerve is
 A. C5 C6
 B. C6 C7
 C. C7 C8
 D. C8 T1
53. Posterior interosseous nerve passes through which osteofascial compartment under extensor retinaculum?
 A. Second
 B. Third
 C. Fourth
 D. Fifth
54. Which of the following muscles of back is *not* supplied by dorsal rami of spinal nerves?
 A. Rhomboideus major
 B. Erector spinae
 C. Semispinalis capitis
 D. Splenius capitis
55. Which of the following structures does *not* form a content of lower part of anterior compartment of arm?
 A. Median nerve
 B. Radial nerve
 C. Ulnar nerve
 D. Brachial artery
56. Which of the following statements is true about coracobrachialis muscle?
 A. It arises in common with long head of biceps
 B. It is pierced by musculocutaneous nerve
 C. It is inserted into coronoid process of ulna
 D. It is medial rotator of arm
57. All the following features occur at the level of insertion of coracobrachialis muscle *except*
 A. ulnar nerve goes to posterior compartment
 B. radial nerve enters the anterior compartment
 C. median nerve crosses brachial artery
 D. basilic vein becomes axillary vein

58. Which of the following statements is *not* true about Flexor digitorum superficialis?
- A. Its synovial sheath is common with that of flexor digitorum profundus
 - B. In front of wrist its tendons are arranged in two strata
 - C. It is supplied by anterior interosseous nerve
 - D. Each of its tendon splits into two before insertion.
59. Which of the following statements is *not* true about Flexor carpi ulnaris?
- A. It is supplied by ulnar nerve
 - B. It is inserted into pisiform bone
 - C. It flexes the little finger
 - D. Pisohamate and pisometacarpal ligaments are continuations of its tendon.
60. Which of the following muscles does *not* help in abduction on wrist joint?
- A. Abductor pollicis longus
 - B. Extensor pollicis brevis
 - C. Extensor carpi radialis longus
 - D. Brachioradialis.
61. Which of the following is *not* a thenar muscle?
- A. Flexor pollicis longus
 - B. Opponens pollicis
 - C. Abductor pollicis brevis
 - D. Flexor pollicis brevis
62. All the following factors add to instability of shoulder joint *except*
- A. shallow glenoid cavity
 - B. large head of humerus relative to glenoid cavity
 - C. laxity of capsule
 - D. intracapsular origin of long head of biceps
63. The labrum glenoidale is
- A. made up of elastic cartilage
 - B. triangular in cross section
 - C. not covered by synovial membrane
 - D. does not add to stability of joint.

Ans. 58. C 59. C 60. D 61. A 62. D 63. B

64. Which of the following muscles initiates abduction on shoulder joint?
- Supraspinatus
 - Deltoid
 - Long head of biceps
 - Serratus anterior
65. Which of the following muscles is a flexor of forearm in midprone position?
- Biceps brachii
 - Brachialis
 - Brachioradialis
 - Pronator teres
66. Which of the following statements regarding carrying angle is true?
- Extended forearm is deviated medially in relation to arm.
 - It is due to long lateral flange of trochlea
 - It disappears in prone position
 - Deviation is less in case of females
67. Which of the following joints is *not* a hinge joint?
- Elbow
 - Metacarpophalangeal
 - Distal Interphalangeal
 - Proximal Interphalangeal
68. Which of the following joints has articular disc?
- Metacarpophalangeal
 - Sternoclavicular
 - Wrist joint
 - Superior radioulnar joint
69. All the following statements are true regarding middle phalanx *except*
- dorsally related to extensor apparatus
 - palmar aspects of its base gives attachment to a slip of palmar aponeurosis
 - flexor digitorum superficialis is inserted on its base.
 - flexor digitorum profundus is related to its palmar surface

Ans. 64. A 65. C 66. C 67. B 68. B 69. B

70. Which of the following arteries arises from second part of axillary artery?
- A. Superior thoracic
 - B. Lateral thoracic
 - C. Subscapular
 - D. Circumflex scapular
71. Brachial artery lies on the anterior aspect of all the following structures *except*
- A. teres major
 - B. triceps
 - C. coracobrachialis
 - D. brachialis
72. All the following are branches of brachial artery *except*
- A. profunda brachii
 - B. nutrient artery to humerus
 - C. ulnar collateral
 - D. common interosseous
73. Superficial palmar arch
- A. lies superficial to palmar aponeurosis
 - B. is accompanied by superficial branch of ulnar nerve
 - C. gives three common digital arteries
 - D. is completed by princeps pollicis artery
74. In relation to axillary vein, which of the following statements is correct?
- A. It is the continuation of basilic vein
 - B. It pierces the clavipectoral fascia
 - C. It terminates into brachiocephalic vein
 - D. It lies lateral to the axillary artery
75. Which of the following arteries accompanies the axillary nerve through the quadrangular space?
- A. Anterior circumflex humeral artery
 - B. Posterior circumflex humeral artery
 - C. Circumflex scapular artery
 - D. Profunda brachii artery
 - E. Suprascapular artery

76. Which of the following is *not* a branch of deep palmar arch?
- Recurrent branches
 - Perforating arteries
 - Palmar digital arteries
 - Palmar metacarpal arteries
77. Which of the following arteries does not participate in scapular anastomosis?
- Suprascapular
 - Transverse cervical
 - Lateral thoracic
 - Circumflex scapular
78. Cephalic vein
- arises from the medial end of dorsal venous arch
 - terminates by joining basilic vein
 - continues as axillary vein at the lower border of teres major
 - joins axillary vein in axilla
 - has none of the above properties
79. Which of the following muscles will be paralysed when recurrent branch of median nerve is injured?
- Abductor pollicis longus
 - Extensor pollicis longus
 - Extensor pollicis brevis
 - Abductor pollicis brevis
 - None of the above
80. Median nerve receives contribution from the
- lateral cord
 - medial cord
 - posterior cord
 - lateral and medial cords
 - medial and posterior cords
81. Intercostobrachial nerve is usually a branch of the
- lower trunk of brachial plexus
 - medial cord
 - first intercostal nerve
 - second intercostal nerve
 - third intercostal nerve

Ans. 76. C 77. C 78. D 79. D 80. D 81. D

82. T

A

B

C

D

83. T

o

A

B

C

D

84.

85.

86.

87.

Ans

82. The nail bed of middle finger is usually supplied by
A. palmar digital branch of ulnar nerve
B. dorsal branch of ulnar nerve
C. palmar digital branch of median nerve
D. radial nerve
83. The nerve most likely to be injured in fractures of the shaft of the humerus is the
A. median
B. musculocutaneous
C. radial
D. axillary
E. ulnar
84. The subacromial bursa
A. communicates with the synovial cavity of shoulder joint
B. lies under the tendon of supraspinatus
C. does not communicate with subdeltoid bursa
D. lies under the coracoacromial ligament
E. has all the above properties
85. The scapula and upper limb are suspended from the clavicle by
A. coracoacromial ligament
B. coracohumeral ligament
C. coracoclavicular ligament
D. costoclavicular ligament
86. Injury to which part of brachial plexus results in "Policeman receiving a tip position" of the upper limb?
A. Lateral cord
B. Upper trunk
C. Lower trunk
D. Middle trunk
87. 'Saturday night paralysis' results from injury to
A. radial nerve
B. ulnar nerve
C. median nerve
D. musculocutaneous nerve

Ans. 82. C 83. C 84. D 85. C 86. B 87. A

88. In injury of radial nerve, which of the following movements will *not* be possible?
- Abduction at metacarpophalangeal joints
 - Adduction at metacarpophalangeal joints
 - Extension at metacarpophalangeal joints
 - Extension at interphalangeal joints.
89. Student's elbow is due to inflammation of
- subcutaneous olecranon bursa
 - medial epicondyle
 - lateral epicondyle
 - skin over olecranon
90. Tennis elbow can result from all the following conditions *except*
- sprain of ulnar collateral ligament
 - tearing of fibres of extensor carpi radialis brevis muscle
 - inflammation of lateral epicondyle
 - inflammation of bursa underneath extensor carpi radialis brevis
91. All the following statements are true regarding fracture of scaphoid *except*
- most frequently fractured carpal bone
 - high incidence of avascular necrosis
 - low incidence of non-union
 - clinically diagnosed by tenderness in anatomic snuffbox.
92. All of the following statements are true about lunate bone *except*
- half moon shaped bone
 - articulates distally with head of capitate
 - usually dislocates dorsally
 - median nerve assessment is important in managing its dislocation
93. All the following statements regarding palmar creases are true *except*
- transverse crease in the distal palm corresponds to metacarpophalangeal joints
 - proximal crease of fingers is located at base of proximal phalanx

Ans. 88. C 89. A 90. A 91. C 92. C

Ans.

- C. middle crease of fingers is at proximal interphalangeal joint
D. distal crease of fingers lies opposite distal interphalangeal joint
94. Symptoms of carpal tunnel syndrome pertain to involvement of
A. flexor digitorum superficialis
B. flexor digitorum profundus
C. ulnar nerve
D. median nerve
95. "Ape" hand deformity results from injury to
A. ulnar nerve
B. radial nerve
C. median nerve
D. first carpometacarpal joint
96. Sensory loss over lateral half of the anterior surface of forearm is due to involvement of which of the following nerves?
A. Median
B. Ulnar
C. Radial
D. Musculocutaneous
97. Ulnar bursa communicates distally with digital synovial sheath of which digit?
A. Little finger
B. Ring finger
C. Middle finger
D. Index finger
98. Fanning of digits is not possible in injury to
A. radial nerve
B. deep branch of ulnar nerve
C. superficial branch of ulnar nerve
D. median nerve.
99. All the following statements are true about midpalmar space except
A. it lies under medial part of palmar aponeurosis
B. it communicates with medial three lumbrical canals
C. it communicates with digital synovial sheaths
D. it is separated from hypothenar muscles by medial palmar septum

Ans. 93. B 94. D 95. C 96. D 97. A 98. B 99. C

groove)
tubercle)

bone)

radial n.)

)
axed)
elbow

100. The clavicle

- A. articulates with the sternum forming cartilaginous joint
- B. forms a fibrous joint with the acromion
- C. is not united to the clavicle of the opposite side
- D. is convex anteriorly in its medial two third part

Directions: For each of the incomplete statements or questions below, one or more completions or answers given is correct, Select

- A. if only 1,2 and 3 are correct;
- B. if only 1 and 3 are correct;
- C. if only 2 and 4 are correct;
- D. if only 4 is correct;
- E. if all are correct.

101. The coracoid process

- 1. is subcutaneous
- 2. has a bursa underneath it
- 3. gives attachment to deltoid
- 4. transmits forces from upper limb to the clavicle

102. The fracture of the surgical neck of the humerus is likely to be associated with

- 1. paralysis of deltoid muscle
- 2. tearing of brachialis muscle
- 3. tearing of the capsule of shoulder joint
- 4. injury to brachial artery

103. The head of ulna

- 1. articulates with ulnar notch of radius
- 2. is separated from the wrist joint by an articular disc
- 3. is separated from the styloid process by a groove in which lies the tendon of extensor carpi ulnaris
- 4. is palpable below the styloid process of ulna

104. The shoulder joint

- 1. has a tight capsule
- 2. is protected all around by deltoid
- 3. communicates with subacromial bursa
- 4. allows only about 120° of abduction

Ans. 100. D 101. D 102. B 103. A 104. D

105. The elbow

- 1. is a
- 2. is a
- 3. is a
- 4. is a

106. The supinator

- 1. is a
- 2. is a
- 3. allows
- 4. is a

107. The wrist

- 1. is for
- 2. is el
- 3. is fle
- digit
- 4. has g

108. True statement

- 1. it art
- 2. it par
- 3. it is u
- 4. it arti

109. The metacarpal

- 1. is a h
- 2. is flex
- 3. is exte
- 4. is abd

110. The interosseous

- 1. has fit
- 2. gives c
- 3. is pier
- 4. transm

111. The axilla

- 1. has a p
- muscle
- 2. has axi
- medial

Ans. 105. E 106. E 107. C 108. A 109. B 110. D 111. C

105. The elbow joint

1. is a hinge variety of synovial joint
2. communicates with the superior radioulnar joint
3. is extended primarily by triceps
4. is protected by strong collateral ligaments

106. The superior radioulnar joint

1. is a pivot type of joint
2. is separated from the elbow joint by annular ligament
3. allows movements of supination and pronation
4. is commonly dislocated in adults

107. The wrist joint

1. is formed between ulna, radius, scaphoid, lunate and triquetral
2. is ellipsoid type of synovial joint
3. is flexed primarily by flexor digitorum superficialis and flexor digitorum profundus
4. has greater range of adduction than abduction

108. True statements concerning the lunate include

1. it articulates with the articular disc
2. it participates in the midcarpal joint
3. it is usually dislocated anteriorly
4. it articulates with the second metacarpal bone

109. The metacarpophalangeal joint of index finger

1. is a hinge type of synovial joint
2. is flexed by first lumbrical muscle
3. is extended primarily by first dorsal interosseous
4. is abducted by first dorsal interosseous

110. The interosseous membrane

1. has fibres directed upwards and medially
2. gives origin to deep muscles of forearm
3. is pierced by posterior interosseous nerve and artery
4. transmits to ulna force directed upwards from hand and radius

111. The axilla

1. has a posterior wall constituted by scapula and subscapularis muscle
2. has axillary lymph nodes which also drain lymph from the medial part of mammary gland

Ans. 105. E 106. B 107. C 108. A 109. C 110. C

lar groove
& lesser tub

groove)

tes radial m

ed)
flexed)
extended
on elbow

3. communicates with neck behind the middle third of clavicle
4. has an anterior wall along which the long thoracic nerve lies

112. The pectoralis major

1. takes origin from the second to sixth ribs
2. is inserted into the lateral lip of bicipital groove
3. is the main adductor and lateral rotator of arm at the shoulder joint
4. is supplied by lateral and medial pectoral nerves

113. The pectoralis minor

1. takes origin from third to fifth ribs
2. is inserted on the tip of acromion process
3. is supplied by medial pectoral nerve
4. protracts the scapula and adducts the arm

114. The serratus anterior

1. is inserted mainly on the superior angle of scapula
2. protracts the scapula and rotates it laterally to produce overhead abduction
3. is supplied by cervical 6, 7 and 8 segments of the spinal cord
4. paralysis results in winging of scapula

115. The deltoid muscle

1. takes origin from the posterior border of the lateral third of clavicle
2. is a multipennate muscle
3. is supplied by a branch from lateral cord of brachial plexus
4. produces abduction, flexion and extension on the shoulder joint

116. The biceps brachii

1. has a long head arising from the coracoid process and a short head arising from supraglenoid tubercle of scapula
2. is supplied by median nerve
3. is attached to radius through bicipital aponeurosis
4. is a strong supinator

117. The brachialis

1. is inserted on radial tuberosity
2. has a double nerve supply

Ans. 111. A 112. C 113. B 114. C 115. C 116. D

3. flexes the arm on shoulder joint
4. flexes the forearm on elbow joint

118. The triceps muscle

1. has a long head arising from supraglenoid tubercle of scapula
2. has a deeper part called articularis cubiti, which is inserted on the capsule of elbow joint
3. receives all its nerve supply from radial nerve in the radial groove
4. is the main extensor of forearm

119. The supinator

1. takes origin from the supinator crest of ulna
2. is pierced by the posterior intersseous nerve
3. is a strong supinator when elbow is kept extended
4. forms the floor of cubital fossa

120. In relations to cubital fossa

1. the median cubital vein runs obliquely across it
2. is bounded laterally by the medial border of pronator teres
3. median nerve leaves the fossa by passing between the two heads of pronator teres
4. pulsations of brachial artery can be felt just lateral to the tendon of biceps brachii

121. The extensor carpi ulnaris

1. is supplied by the ulnar nerve
2. is inserted into the pisiform bone
3. passes superficial to the extensor retinaculum
4. produces extension and adduction of hand

122. The flexor carpi radialis

1. is inserted into the bases of second and third metacarpals
2. is supplied by radial nerve
3. has its own synovial sheath
4. produces flexion of second and third fingers

123. The brachioradialis muscle

1. takes origin from the lower one third of the lateral supracondylar ridge of humerus
2. is supplied by the radial nerve

Ans. 117. C 118. C 119. E 120. B 121. D 122. B

3. flexes the hand on the wrist joint
4. brings the forearm in midprone position and flexes it

124. The flexor digitorum profundus

1. is supplied by both ulnar and median nerves
2. tendons are arranged in two layers in front of the wrist joint
3. has a synovial sheath in common with the flexor digitorum superficialis
4. is the prime flexor of proximal interphalangeal joint

125. All the superficial flexor muscles of the forearm

1. arise by a common tendon from the medial epicondyle of humerus
2. are supplied by the median nerve
3. are flexors of the elbow joint
4. enter the hand deep to flexor retinaculum

126. The lumbrical muscles

1. arise from the tendons of flexor digitorum superficialis
2. are supplied by digital branches of ulnar and median nerves
3. pass along the ulnar side of each digit to reach dorsal digital expansion
4. flex the metacarpophalangeal joints and extend the interphalangeal joints

127. The dorsal interossei

1. are bipennate muscles
2. are supplied by the radial nerve
3. are abductors of digits
4. are extensors of metacarpophalangeal joints

128. The thenar space

1. contains thenar muscles
2. is separated from midpalmar space by a septum
3. communicates with the radial bursa
4. is bounded posteriorly by fascia covering the adductor pollicis muscle

129. The palmar aponeurosis

1. divides into five slips distally
2. gives insertion to palmaris longus

Ans. 123. C 124. B 125. B 126. D 127. B 128. C

3. undergoes contraction in Volkmann's ischemic contracture
4. is related to superficial palmar arch on its deeper aspect

130. The digital synovial sheath

1. encloses only the tendon of flexor digitorum superficialis
2. extends upto the tip of the terminal phalanx
3. sheath of index finger is continuous with the ulnar bursa
4. forms vincula longa and breva to carry blood vessels to the tendons

131. The extensor expansion

1. is mainly formed by the tendon of extensor digitorum longus
2. is formed dorsal to the proximal interphalangeal joint
3. gives insertion to lumbricals and interossei
4. divides into three parts; a median part attached to the base of the terminal phalanx and two collateral parts attached to the base of the middle phalanx

132. The flexor retinaculum

1. is attached medially to the pisiform and hook of hamate
2. forms carpal tunnel through which the median and ulnar nerves pass
3. prevents buckling of tendons
4. is related superficially to the ulnar and radial arteries

133. The lateral cord of brachial plexus

1. contains fibres from the fifth, sixth and seventh cervical nerves
2. lies lateral to the axillary artery throughout
3. gives contribution to lateral pectoral, musculocutaneous and median nerves
4. is formed from the dorsal divisions of upper and middle trunks

134. The posterior cord of brachial plexus

1. is formed from the posterior divisions of all the trunks
2. lies posterior to the first and second parts of axillary artery
3. supplies mainly the muscles on the posterior aspect of upper limb
4. supplies teres major through thoracodorsal nerve

135. The ulnar nerve

1. arises from the lateral cord

Ans. 129. C 130. D 131. A 132. B 133. A 134. B

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2. passes posterior to medial epicondyle of humerus
3. supplies most of the muscles of front of forearm
4. supplies all the interossei muscles

136. The median nerve

1. supplies cutaneous branches to lateral three and a half digits
2. supplies all short muscles acting on thumb
3. injury leads to ape-hand deformity
4. injury results in loss of adduction and abduction of extended fingers

137. The median nerve

1. arises from the posterior and lateral cords of brachial plexus
2. supplies muscles of front of arm
3. lies just medial to the tendon of flexor digitorum superficialis
4. passes through the carpal tunnel

138. The ulnar nerve

1. lies medial to the upper part of brachial artery
2. is accompanied by inferior ulnar collateral artery in the back of arm
3. enters the forearm between the two heads of flexor carpi ulnaris
4. supplies shoulder joint

139. The radial nerve

1. is the largest branch of the posterior cord of brachial plexus
2. gives no branch in the axilla
3. runs in the radial groove between lateral and medial heads of triceps
4. injury results in Klumpke's paralysis

140. The radial nerve supplies

1. triceps
2. anconeus
3. brachialis
4. skin over lower half of deltoid muscle

141. The ulnar nerve supplies

1. flexor carpi ulnaris
2. extensor carpi ulnaris
3. first dorsal interosseous
4. first lumbrical

Ans. 135. C 136. B 137. D 138. B 139. B 140. A 141. B

Chapter 2

142. The

- 1.
- 2.
- 3.
- 4.

143. The

- 1.
- 2.
- 3.
- 4.

144. The

- 1.

- 2.

- 3.

- 4.

145. The

- 1.

- 2.

- 3.

- 4.

146. The

- 1.

- 2.

- 3.

- 4.

147. The

- 1.

- 2.

- 3.

- 4.

Ans. 142.

142. The anterior interosseous nerve supplies

1. flexor digitorum superficialis
2. flexor digitorum profundus
3. pronator teres
4. flexor pollicis longus

143. The musculocutaneous nerve

1. is a branch of the lateral cord of brachial plexus
2. passes between the two heads of biceps brachii
3. supplies all muscles of anterior compartment of arm
4. continues as medial cutaneous nerve of forearm

144. The axillary artery

1. continues as brachial artery at the lower border of pectoralis major
2. lies medial to the axillary vein
3. gives lateral thoracic artery as its first branch
4. is related throughout to brachial plexus and its branches

145. The brachial artery

1. is the continuation of axillary artery at the upper border of teres major
2. is crossed by median nerve in the middle of arm
3. can be felt pulsating just lateral to the tendon of biceps brachii
4. is used for measuring blood pressure

146. The radial artery

1. is used for feeling pulse
2. enters the palm between the two heads of first dorsal interosseous muscle
3. gives princeps pollicis in the palm
4. is the main artery forming deep palmar arch

147. The posterior interosseous artery usually

1. enters the posterior compartment of forearm in company with the posterior interosseous nerve
2. passes under the extensor retinaculum along with the extensor digitorum muscle
3. is a branch of radial artery
4. gives interosseous recurrent artery which anastomosis with middle collateral branch of profunda brachii artery

Ans. 142. C 143. B 144. D 145. C 146. E 147. D

148. The basilic vein

1. arises from veins on the palmar aspect of hand
2. pours most of its blood into cephalic vein through median cubital vein
3. terminates by joining venae comitantes of brachial artery in the middle of arm
4. continues as axillary vein at the lower border of teres major

149. Which of the following statements are true for lymphatic drainage of the breast?

1. The axillary lymph nodes receive 75 percent of the lymph from the breast
2. The parasternal lymph nodes receive lymph from both the medial and lateral halves of the breast
3. Cancer cells from one breast can travel to the other breast through lymphatics
4. Some lymphatics draining the upper part of mammary gland parenchyma directly go to the inferior deep cervical lymph nodes

Directions: Each question consists of an assertion and a reason. Responses should be chosen as follows:

- A. if the assertion and reason are true statements and the reason is a correct explanation of the assertion;
- B. if the assertion and reason are true statements but the reason is not a correct explanation of the assertion;
- C. if the assertion is true but the reason is a false statement;
- D. if the assertion is false but the reason is a true statement;
- E. if both assertion and reason are false statements.

150. Fracture of the middle of the shaft of humerus results in a complete paralysis of triceps muscle

Because
the radial nerve gets injured

151. The dislocation of shoulder joint occurs primarily in the anterior direction

Because
the anterior part of capsule has an opening through which the joint cavity communicates with the subscapular bursa

Ans. 148. D 149. E 150. D 151. D

- 152. The commonest site for fracture of clavicle is at the junction of its lateral 2/3 and medial 1/3**
Because
a strong coracoclavicular ligament is attached to the lateral 1/3 of the clavicle
- 153. Injury to the upper trunk of brachial plexus results in impairment of abduction at shoulder joint**
Because
it results in paralysis of deltoid and supraspinatus muscles
- 154. Injury to the long thoracic nerve results in winging of scapula**
Because
scapula is retracted by the rhomboideus major and minor muscles
- 155. Intramuscular injection in deltoid muscle should be given into its lower half**
Because
upper half of the muscle protects the shoulder joint
- 156. When the forearm is extended and the hand supinated, its long axis is directed somewhat laterally making an angle of about 165° with the long axis of arm**
Because
the medial epicondyle of humerus is larger than the lateral epicondyle
- 157. Dislocation of the elbow joint is frequently associated with fracture of upper end of ulna**
Because
whole of the olecranon process of ulna is intracapsular
- 158. In small children, subluxation of the head of radius is a common occurrence**
Because
the head of radius is very small at this stage
- 159. On the wrist joint, the range of abduction is more as compared to the range of adduction**
Because
there are more number of abductors than adductors acting on the wrist joint

Ans. 152. D 153. A 154. B 155. B 156. B 157. C 158. A 159. D

160. Fracture of scaphoid is often followed by avascular necrosis of its proximal fragment
Because radial artery is always injured simultaneously
161. Lunate is the commonest carpal bone to get dislocated
Because it takes part in the formation of the wrist joint as well as the midcarpal joint
162. Injury to the ulnar nerve in the arm does not result in claw hand
Because the interossei and lumbricals are supplied by deep branch of ulnar nerve which arises in the palm
163. Injury to the superficial branch of radial nerve affects power to supinate
Because the supinator and brachioradialis are paralysed
164. Dislocation of lunate can result in loss of adduction of thumb
Because median nerve is pressed against the flexor retinaculum
165. Bleeding from the deep palmar arch can be stopped by ligating ulnar artery
Because the deep palmar arch is mainly formed by the ulnar artery
166. An infection of the synovial sheath within the fifth digit can spread into the fourth digit
Because the fourth and the fifth digits lie close to each other
167. Injury to the recurrent branch of median nerve may result in loss of opposition of thumb
Because the palmar aspects of thumb and index finger will be anaesthetised

Ans. 160.C 161.B 162.D 163.E 164.D 165.E 166.D 167.C

168. **Axillary lymph nodes may get involved in cancer of the axillary tail of mammary gland by direct spread**
Because
the axillary tail is deep to the deep fascia and lies in close proximity to the axillary lymph nodes
169. **Basilic vein is more often used for cardiac catheterisation than the cephalic vein**
Because
most of the blood from cephalic vein is usually carried to basilic vein by the median cubital vein
170. **The median cubital vein is commonly used for blood sampling and intravenous injections**
Because
it is superficial and relatively fixed due to its communication with deep veins
171. **Injury to ulnar nerve in arm does *not* result in any cutaneous sensory loss**
Because
it does not give any branch in the arm
172. **Thecal whitlows of little finger may spread to above the wrist**
Because
the digital synovial sheath for the little finger communicates with the ulnar bursa
173. **In the case of the fracture of radius below the insertion of pronator teres the proximal fragment is pronated**
Because
the action of pronator teres is stronger than the supination action of the biceps brachii
174. **In amputation of the arm, the brachialis muscle is likely to retract more than the biceps brachii**
Because
biceps has two heads while brachialis has only one
175. **Injury to the distal epiphysial line of the humerus in a growing child will give rise to a considerable degree of shortening**

Ans. 168. A 169. D 170. A 171. D 172. A 173. E 174. B

Because
the lower end of the humerus is the growing end

176. The swelling of hand due to edema or extravasation of blood is apparent on the dorsal aspect rather than the palmar aspect
Because
the subcutaneous tissue is lax on the dorsal aspect

177. Ligation of brachial artery below the origin of profunda brachii may not impair circulation through forearm
Because
there is a rich anastomosis around elbow joint

178. Injury to the median nerve in the middle of forearm will lead to complete loss of flexion of wrist
Because
most of the muscles of front of forearm are supplied by the median nerve

179. In a case of injury to the deep branch of ulnar nerve the patient will not be able to hold paper tightly between his fingers
Because
the dorsal interossei will be paralysed

180. An inflammatory swelling of the tendon sheath may lead to necrosis of the tendons
Because
the swelling is likely to compress the vincular blood vessels which supply the tendons.

Ans. 175. E 176. A 177. A 178. D 179. B 180. A

The Lower Limb

Directions: Each of the incomplete statements or questions below is followed by four or five suggested completions or answers. Select the one which is BEST in each case.

1. The acetabulum

- A. is directed backwards, downward and laterally
- B. is completely articular
- C. is formed by ilium and pubis only
- D. is deepened by labrum acetabulare
- E. shows a discontinuity in the superior rim, the acetabular notch

2. The ischiopubic rami fuse around the age of

- A. 1 year
- B. 7 year
- C. 14 year
- D. 21 year

3. Which of the following structures does *not* pass through the lesser sciatic foramen?

- A. Internal pudendal vessels
- B. Tendon of obturator externus
- C. Pudendal nerve
- D. Nerve to obturator internus

4. Which of the following is the most important feature of the female pelvis?

- A. Wide and deep sciatic notch
- B. Heart shaped inlet of pelvis
- C. Subpubic angle of 90° or more
- D. Triangular obturator foramen
- E. Ischiopubic ramus is everted

Ans. 1. D 2. B 3. B 4. C

5. The neck of the femur
- is completely intracapsular
 - is completely extracapsular
 - has medial half intracapsular and lateral half extracapsular
 - has anterior surface intracapsular and posterior surface extracapsular
 - is mostly intracapsular except for its posterolateral part which is extracapsular
6. Which of the following statements is *not* true for medial condyle of femur?
- It has a larger tibial articular surface than the lateral condyle
 - The tibial articular surface is curved
 - It bears a groove on its medial surface for the origin of popliteus
 - It is separated from the lateral condyle posteriorly by intercondylar notch
7. The lower epiphyseal line of femur passes
- above the adductor tubercle
 - through the adductor tubercle
 - below the adductor tubercle
 - through the epicondyles of femur
8. The adductor tubercle gives attachment to
- adductor longus
 - adductor brevis
 - adductor magnus
 - medial head of gastrocnemius
 - tibial collateral ligament
9. The medial surface of tibia
- is mostly subcutaneous
 - gives origin to tibialis anterior
 - gives insertion to semimembranosus
 - gives origin to sartorius, gracilis and semitendinosus
 - has none of the above properties
10. The bone forming the heel is
- lower end of tibia
 - lower end of fibula
 - calcaneum
 - talus

Ans. 5. E 6. C 7. B 8. C 9. A 10. C

11. W

lo

A

B

C

D

E

12. V

A

B

C

D

E

13.

11. Which of the following bones does *not* take part in the medial longitudinal arch?
- A. Calcaneum
 - B. Talus
 - C. Cuboid
 - D. Navicular
 - E. Lateral cuneiform
12. Which of the following structures lies in the sinus tarsi?
- A. Tendon of tibialis posterior
 - B. Dorsalis pedis artery
 - C. Deep peroneal nerve
 - D. Interosseous talocalcaneal ligament
 - E. Bifurcate ligament
13. The patella
- A. is a sesamoid bone which articulates with the tibia
 - B. has a rough posterior surface
 - C. has an apex to which ligamentum patellae is attached
 - D. has a tendency for medial dislocation
 - E. is surrounded on all sides by the tendon of quadriceps femoris
14. Which of the following is *not* the growing end?
- A. Lower end of femur
 - B. Upper end of tibia
 - C. Lower end of fibula
 - D. Proximal end of first metatarsal
15. The bones transmitting body weight include all the following *except*
- A. femur
 - B. tibia
 - C. fibula
 - D. talus
 - E. calcaneum
16. Which of the following muscles is attached to lesser trochanter?
- A. Pectineus
 - B. Iliopsoas
 - C. Adductor longus
 - D. Adductor brevis

Ans. 11. C 12. D 13. C 14. C 15. C 16. B

24.

25.

26.

27.

28.

29.

17. Which of the following structures is *not* attached to linea aspera?
- Gluteus maximus
 - Adductor longus
 - Vastus lateralis
 - Vastus medialis
18. All the following statements are true regarding distal end of femur *except*
- it is the growing end
 - ossification centre appears just before birth
 - presence of ossification center has medicolegal importance
 - articulates with tibia and fibula
19. All the following muscles are attached to anterior of fibula *except*
- tibialis anterior
 - extensor digitorum longus
 - extensor hallucis longus
 - peroneus tertius
20. Which of the following muscles is attached to head of fibula?
- Lateral head of gastrocnemius
 - Plantaris
 - Biceps femoris
 - Tensor fasciae latae
21. Which of the following muscles is *not* attached to calcaneus?
- Soleus
 - Plantaris
 - Flexor digitorum longus
 - Flexor digitorum accessorius
22. How much weight is transmitted to the ground by calcaneus?
- 10%
 - 25%
 - 50%
 - 75%
23. Which of the following ligaments limits adduction on hip joint?
- Ischiofemoral

Ans. 17. A 18. D 19. A 20. C 21. C 22. C

Ans.

- B. Transverse acetabular
C. Ligamentum teres
D. Labrum acetabulare
24. **Anterior dislocation of tibia on femur is prevented by**
A. ligamentum patellae
B. anterior cruciate ligament
C. posterior cruciate ligament
D. oblique popliteal ligament
25. **Rotatory movement on knee joint occurs**
A. between femur and menisci
B. between tibia and menisci
C. in both upper and lower compartments
D. at patello-femoral joint
26. **Which of the following muscles acts on knee as well as ankle joint?**
A. Tibialis anterior
B. Gastrocnemius
C. Soleus
D. Popliteus
27. **In coxa vera**
A. the neck-shaft angle of femur is increased
B. the neck-shaft angle of femur is decreased
C. the neck-shaft angle is unchanged
D. the outer angle between thigh and leg is increased
28. **In telepes-equinovarus deformity, the foot is**
A. plantar flexed
B. adducted
C. inverted
D. shows all these features
E. shows none of these features
29. **When the foot is on the ground, the terminal phase of extension is associated with**
A. medial rotation of femur
B. lateral rotation of femur
C. medial rotation of tibia
D. lateral rotation of tibia

Ans. 23. C 24. B 25. B 26. B 27. B 28. D 29. A

30. In the erect posture the line of gravity falls
- in front of the hip joint
 - behind the knee joint
 - in front of the ankle joint
 - in front of the hip, knee and ankle
 - behind the hip, knee and ankle
31. The attachments of the tibial collateral ligament do *not* include
- medial epicondyle of femur
 - medial condyle of tibia
 - medial surface of the shaft of tibia
 - medial intercondylar tubercle of tibia
 - medial meniscus
32. The medial meniscus is attached to the
- femur by menisiofemoral ligament
 - tendon of popliteus
 - anterior cruciate ligament
 - posterior cruciate ligament
 - none of the above
33. The suprapatellar bursa
- communicates with the knee joint occasionally
 - lies between the quadriceps femoris muscle and anterior surface of femur
 - is often inflamed in housemaids
 - is pulled up by articularis genu during flexion of leg on the knee joint
34. The superior tibio-fibular joint
- is a fibrous joint
 - is a synovial joint
 - communicates with the knee joint
 - allows no movement at all
35. Which of the following is intracapsular intrasynovial?
- Ligamentum teres of femur
 - Cruciate ligaments
 - Lateral and medial menisci
 - Interosseous talocalcaneal ligament
 - Transverse acetabular ligament

Ans. 30. C 31. D 32. E 33. B 34. B 35. C

36. The
- A.
 - B.
 - C.
 - D.
37. The
- A.
 - B.
 - C.
 - D.
38. The
- A.
 - B.
 - C.
 - D.
 - E.
39. The
- A.
 - B.
 - C.
 - D.
 - E.
40. The
- A.
 - B.
 - C.
 - D.
 - E.
- 41.

36. **The inferior tibio-fibular joint**
A. communicates with the ankle joint
B. is between the lateral and medial malleoli
C. is a fibrous joint
D. allows no movement at all
37. **The ankle joint is most stable in**
A. plantar flexion
B. dorsi flexion
C. Inversion
D. Eversion
38. **The quadriceps femoris is**
A. extensor of thigh
B. extensor of leg
C. inserted into the apex of patella
D. the unlocking muscle on the knee joint
E. supplied by sciatic nerve
39. **The gluteus medius muscle**
A. is supplied by the inferior gluteal nerve
B. is inserted on the anterior aspect of greater trochanter
C. is extensor of the thigh
D. raises the pelvis of opposite side while walking
E. is completely undercover of gluteus maximus
40. **The biceps femoris muscle**
A. arises from hip bone only
B. is inserted on the neck of fibula
C. is closely related to the common peroneal nerve in popliteal fossa
D. flexes the thigh and leg
E. is supplied only by tibial part of sciatic nerve
41. **Which of the following muscles is extensor of both the hip and knee joints?**
A. Hamstrings
B. Quadriceps femoris
C. Tensor fascia latae
D. Gluteus medius
E. None of the above

Ans. 36. C 37. B 38. B 39. D 40. C 41. E

42. Which of the following muscles is flexor of both the hip and knee joints?
- Biceps femoris
 - Rectus femoris
 - Sartorius
 - Gracilis
 - None of the above
43. Which of the following muscles produces only lateral rotation of thigh?
- Psoas major
 - Adductor longus
 - Quadratus femoris
 - Gluteus maximus
44. The boundaries of the femoral triangle are
- inguinal ligament, sartorius and pectineus
 - inguinal ligament, psoas major and adductor longus
 - adductor longus, sartorius and inguinal ligament
 - iliacus, sartorius and inguinal ligament
45. The structures in the femoral sheath are arranged from lateral to medial side as
- femoral nerve, femoral artery, femoral vein
 - femoral artery, femoral vein, femoral canal
 - femoral vein, femoral artery, femoral canal
 - femoral vein, femoral artery, femoral nerve
46. Medial boundary of femoral ring is formed by
- inguinal ligament
 - lacunar ligament
 - pectineal ligament
 - femoral vein
47. All the following features pertain to gracilis muscle except
- it is attached to pubic bone
 - it is supplied by obturator nerve
 - it is adductor of thigh
 - it extends the knee joint

48. **Tensor fasciae latae**
A. arises from anterior superior iliac spine
B. inserted into iliotibial tract
C. supplied by inferior gluteal nerve
D. causes flexion on knee joint
49. **Adductor magnus muscle**
A. is supplied by obturator and femoral nerves
B. arises from pubic bone and ischial tuberosity
C. is inserted into lateral lip of linea aspera
D. causes flexion and adduction of thigh
50. **Obturator externus muscle has all the following features except**
A. arises from obturator membrane
B. inserted into trochanteric fossa
C. supplied by obturator nerve
D. produces medial rotation of thigh
51. **All the following structures pass under the flexor retinaculum except**
A. tibialis anterior
B. flexor hallucis longus
C. flexor digitorum longus
D. tibial nerve
52. **Which of the following muscles does *not* lie in the anterior compartment of leg?**
A. Extensor hallucis longus
B. Extensor digitorum longus
C. Tibialis anterior
D. Peroneus brevis
53. **The tendon of which of the following muscles crosses the sole from lateral to medial side?**
A. Flexor digitorum longus
B. Peroneus longus
C. Tibialis posterior
D. Flexor hallucis longus

Ans. 48. B 49. B 50. D 51. A 52. D 53. B

54. At the apex of femoral triangle the structures are arranged from anterior to posterior side as
- femoral vein, femoral artery, profunda femoris artery and profunda femoris vein
 - femoral artery, femoral vein, profunda femoris artery and profunda femoris vein
 - femoral vein, femoral artery, profunda femoris vein and profunda femoris artery
 - femoral artery, femoral vein, profunda femoris vein and profunda femoris artery
55. The articularis genu is supplied by
- nerve to vastus lateralis
 - nerve to vastus medialis
 - nerve to vastus intermedius
 - saphenous nerve
 - obturator nerve
56. Which of the following structures passes through the stem of the inferior extensor retinaculum?
- Tibialis anterior
 - Extensor digitorum longus
 - Extensor hallucis longus
 - Deep peroneal nerve
 - Dorsalis pedis artery
57. The extensor hallucis brevis
- takes origin from talus
 - is inserted into the base of the distal phalanx of big toe
 - is supplied by superficial peroneal nerve
 - extends the interphalangeal joint of big toe
 - has none of the above properties
58. When the dorsal interosseous muscle between the fourth and fifth toes contracts, it results in
- extension of the metatarsophalangeal joint of the fourth toe
 - abduction of the fifth toe
 - adduction of the fifth toe
 - abduction of the fourth toe
 - flexion of the interphalangeal joint of the fourth toe

Ans. 54. D 55. C 56. B 57. E 58. D

59. The movement of inversion and eversion occurs on the
- ankle joint
 - ankle and tarsometatarsal joints
 - subtalar joint
 - subtalar and calcaneo-cuboid joints
 - subtalar and transverse tarsal joints
60. Which of the following evertors is inserted into medial cuneiform and first metatarsal bone?
- Peroneus longus
 - Peroneus brevis
 - Peroneus tertius
 - Tibialis anterior
61. Which of the following invertors is inserted on almost all the tarsal bones *except* talus?
- Tibialis anterior
 - Tibialis posterior
 - Flexor hallucis longus
 - Flexor digitorum longus
62. The main plantar flexor on the ankle joint is
- tibialis posterior
 - flexor digitorum longus
 - gastrocnemius
 - peroneus longus
 - plantaris
63. The femoral nerve (or its branches)
- supplies skin on the medial side of leg and foot
 - supplies skin on the lateral side of thigh
 - supplies quadratus femoris
 - arises from the L3-L5 nerves
 - lies within the femoral sheath
64. The obturator nerve
- arises from the sacral plexus
 - is the sole nerve supply to pectineus
 - enters the thigh through obturator canal
 - injury will mainly interfere with flexion on the hip joint
 - is purely a motor nerve

Ans. 59. E 60. A 61. B 62. C 63. A 64. C

65. Which of the following cutaneous nerves is *not* a branch of femoral nerve?
- Medial cutaneous nerve of thigh
 - Intermediate cutaneous nerve of thigh
 - Lateral cutaneous nerve of thigh
 - Saphenous nerve
66. Area of skin over upper part of femoral triangle is supplied by
- ilioinguinal nerve
 - femoral branch of genitofemoral
 - intermediate cutaneous branch of femoral
 - obturator nerve
67. Accessory obturator nerve
- runs along lateral pelvic wall
 - emerges from obturator foramen
 - supplies pectineus
 - arises from L1 and L2 nerves
68. Inferior gluteal nerve supplies
- gluteus minimus
 - gluteus medius
 - gluteus maximus
 - skin over lower part of gluteal region
69. Muscles of anterior compartment of leg are supplied by
- superficial peroneal nerve
 - deep peroneal nerve
 - common peroneal nerve
 - saphenous nerve
70. Superior gluteal nerve
- has root value L4, L5, S1
 - enters gluteal region through lesser sciatic foramen
 - runs between gluteus maximus and gluteus medius
 - supplies all three gluteal muscles
71. Muscles of posterior compartment of leg are supplied by
- common peroneal nerve
 - tibial nerve

Ans. 65. C 66. B 67. C 68. C 69. B 70. A

- C. sural nerve
D. all the above
72. **L4 segment of spinal cord supplies skin of**
A. antero-lateral aspect of thigh
B. medial aspect of leg
C. lateral part of dorsum of foot
D. back of thigh
73. **Tibial nerve**
A. courses vertically down the popliteal fossa
B. is covered throughout by skin and fascia only
C. is accompanied by peroneal artery
D. divides into plantar nerves behind the lateral malleolus
74. **The deep peroneal nerve**
A. arises at the level of head of the fibula
B. enters the anterior compartment of leg by piercing the interosseous membrane
C. is purely a motor nerve
D. has a terminal lateral branch ending in a pseudoganglion
E. lies along the medial border of biceps femoris in popliteal fossa
75. **Flexion on the knee joint, after injury to the tibial part of the sciatic nerve, is done by**
A. gastrocnemius
B. popliteus
C. long head of biceps femoris
D. sartorius
E. none of the above
76. **Foot drop is due to injury to the**
A. superficial peroneal nerve
B. deep peroneal nerve
C. tibial nerve
D. medial and lateral plantar nerves
E. tendocalcaneus
77. **The skin on the dorsum of the foot is supplied by**
A. saphenous nerve
B. superficial peroneal nerve

Ans. 71. B 72. B 73. A 74. D 75. D 76. B

- C. deep peroneal nerve
 - D. medial and lateral plantar nerves
 - E. all the above
78. **Intramuscular injection in the gluteal region should be given in**
- A. upper medial quadrant
 - B. lower medial quadrant
 - C. upper lateral quadrant
 - D. lower lateral quadrant
 - E. its centre
79. **The medial plantar nerve supplies**
- A. flexor digitorum brevis
 - B. flexor digitorum accessorius
 - C. second lumbrical
 - D. adductor hallucis
 - E. all the above
80. **The femoral artery**
- A. is the continuation of internal iliac artery
 - B. lies lateral to the femoral nerve
 - C. gives profunda femoris artery
 - D. leaves the femoral triangle by passing behind the adductor longus muscle
 - E. supplies only the lower limb
81. **The profunda femoris artery**
- A. arises from femoral artery in adductor canal
 - B. is the main artery for medial and posterior compartments of thigh
 - C. leaves the adductor canal through hiatus magnus
 - D. gives descending genicular artery
 - E. terminates by dividing into lateral and medial circumflex arteries
82. **The posterior tibial artery**
- A. begins at the upper border of popliteus muscle
 - B. gives peroneal artery which pierces the interosseous membrane
 - C. divides into medial and lateral plantar arteries under the flexor retinaculum

- D. gives nutrient artery to fibula
E. lies deep to the tibialis posterior muscle
83. The pulsations of dorsalis pedis artery can be felt
A. over the medial malleolus
B. over the lateral malleolus
C. at the distal end of first intermetatarsal space
D. on the dorsum of foot lateral to the tendon of extensor hallucis longus
84. The dorsalis pedis artery most commonly arises from
A. peroneal artery
B. anterior tibial artery
C. posterior tibial artery
D. medial plantar artery
E. lateral plantar artery
85. The branches of anterior tibial artery do *not* include the
A. posterior tibial recurrent
B. anterior tibial recurrent
C. peroneal
D. lateral malleolar
E. medial malleolar
86. The great saphenous vein
A. begins on the lateral side of foot
B. ascends behind the medial malleolus
C. pierces the cribriform fascia just medial to pubic tubercle
D. is connected with deep veins through perforating veins
E. drains only the lower limb
87. The perforating veins
A. connect the great saphenous vein with the short saphenous vein
B. connect the superficial veins with the deep veins
C. normally allow passage of blood in both directions
D. normally allow passage of blood from deep to superficial veins
E. are mainly present in the foot
88. The commonest site for venesection of the great saphenous vein is
A. at the saphenous opening

Ans. 82. C 83. D 84. B 85. C 86. D 87. B

- B. anterior to the medial malleolus
 C. posterior to the lateral malleolus
 D. posteromedial to the knee joint
 E. roof of popliteal fossa
89. All the following are branches of femoral artery except
 A. superficial epigastric
 B. deep circumflex iliac
 C. profunda femoris
 D. descending genicular
90. The obturator artery is the branch of
 A. femoral artery
 B. external iliac artery
 C. internal iliac artery
 D. profunda femoris artery
91. The popliteal artery pulse is palpated easily when
 A. leg and thigh are extended
 B. leg is semiflexed
 C. thigh is medially rotated
 D. thigh is laterally rotated
92. Which of the following vessels is used for measuring blood pressure in lower limb?
 A. Femoral
 B. Profunda femoris
 C. Popliteal
 D. Posterior tibial
93. When the back of knee joint is suddenly tapped in a standing person, he tends to fall down. Which of the following muscles is responsible for this phenomenon?
 A. Quadriceps femoris
 B. Biceps femoris
 C. Sartorius
 D. Popliteus
94. Patella has inherent tendency to dislocate laterally. The natural mechanism to prevent this is
 A. long axis of femur is directed downward and medially

- B. forward projection of medial condyle of femur
- C. attachment of vastus medialis on patella extends more distally
- D. larger articular area for patella on medial condyle of femur.

95. All the following statements regarding fracture of tibia are true *except*

- A. common site is the junction of middle and lower thirds of tibia, being the narrowest part
- B. fractures are usually open type as most of the medial surface is subcutaneous
- C. delayed union is common because of less periosteal blood supply
- D. splint-like action of intact fibula prevents proximal and distal fragments of tibia to come in opposition.

96. Which nerve is likely to be injured in posterior dislocation of hip joint?

- A. Superior gluteal
- B. Perforating cutaneous
- C. Pudendal
- D. Sciatic

97. Housemaid's knee is the inflammation of

- A. suprapatellar bursa
- B. prepatellar bursa
- C. infrapatellar bursa
- D. synovial membrane of knee joint

98. Weaver's bottom is the inflammation of bursa between

- A. gluteus maximus and ischial tuberosity
- B. gluteus maximus and greater trochanter
- C. gluteus maximus and vastus lateralis
- D. adductor magnus and ischial tuberosity

99. Which muscle is termed peripheral heart because of valveless veins present in that muscle?

- A. Gastrocnemius
- B. Soleus
- C. Gluteus maximus
- D. Gluteus medius

Ans. 94. C 95. A 96. D 97. B 98. A 99. B

90. MCQs in Anatomy

100. Inability to extend the knee may be due to injury to
A. tibial nerve
B. sciatic nerve
C. femoral nerve
D. obturator nerve
101. Enlarging the femoral ring medially to relieve strangulation of femoral hernia is sometimes met with serious haemorrhage due to rupture of
A. femoral vein
B. femoral artery
C. accessory obturator artery
D. obturator artery
102. All the following features pertain to varicose veins *except* that these veins are
A. tortuous
B. thickened
C. narrowed
D. visible
103. All the following features point to injury to tibial nerve *except*
A. foot is held dorsiflexed and everted
B. inability to stand on toes
C. loss of sensation on sole of foot
D. loss of sensation in back of knee
104. Complete injury of sciatic nerve results in loss of all cutaneous sensibility below the knee *except* along medial side of leg and foot up to ball of big toe. This is due to intact
A. sural nerve
B. saphenous nerve
C. superficial peroneal nerve
D. obturator nerve
105. Which of the following muscles will escape paralysis if sciatic nerve is injured at middle level of thigh?
A. Hamstrings
B. Popliteus
C. Gastrocnemius
D. Soleus

Ans. 100. C 101. C 102. C 103. D 104. B 105. A

Chapter

Direction
or more

106. T

107. T

108. T

109. T

110. T

111. T

Ans.

Direction For each of the incomplete statements or questions below, one or more completions or answers given is correct, Select

- A. if only 2 and 3 are correct;
- B. if only 1 and 3 are correct;
- C. if only 2 and 4 are correct;
- D. if only 4 is correct;
- E. if all are correct.

106. The muscles attached to the greater trochanter include

- 1. obturator externus
- 2. gluteus minimus
- 3. gluteus medius
- 4. gluteus maximus

107. The head of femur

- 1. is completely intracapsular
- 2. forms half the sphere
- 3. gives attachment to ligamentum teres of femur
- 4. is pierced by numerous vascular foramina

108. The neck of fibula

- 1. is easily palpable
- 2. is laterally related to anterior tibial artery
- 3. gives insertion to biceps femoris
- 4. when fractured, is often associated with foot drop

109. The lateral longitudinal arch is maintained by

- 1. peroneus longus
- 2. peroneus brevis
- 3. long and short plantar ligaments
- 4. small muscles of little toe

110. The medial longitudinal arch

- 1. is higher than the lateral arch
- 2. is less resilient than the lateral arch
- 3. is maintained by tibialis posterior
- 4. is flattened in pes cavus

111. The transverse tarsal joints

- 1. include talonavicular and calcaneocuboid joints
- 2. are fibrous joints

Ans. 106. A 107. B 108. D 109. E 110. B

3. allow movements of inversion and eversion
4. allow movements of dorsi flexion and plantar flexion

112. The fibular collateral ligament of knee joint

1. is band shaped
2. splits the tendon of biceps femoris into two parts
3. is attached to the lateral meniscus
4. is taut in extension

113. Which of the following statements is/are true for the hip joint?

1. It is a very stable, ball and socket type of synovial joint
2. Its main flexor is the iliopsoas muscle
3. Extension is limited by the iliofemoral ligament
4. It dislocates mainly anteriorly

114. In relation to the knee joint

1. it is a pure hinge type of synovial joint
2. its proximal and distal articular surfaces are congruent
3. it has no communications
4. the capsule is replaced anteriorly by the patellar retinacula, the patella and the ligamentum patellae

115. In relation to the knee joint

1. The joint cavity is divided into upper and lower compartments by the menisci
2. The movement of rotation occurs in the lower compartment
3. The collateral ligaments become taut in extension and so prevent adduction and abduction
4. The anterior cruciate ligament is attached to the lateral condyle of femur

116. In relation to the ankle joint

1. it is formed by tibia, fibula, talus and calcaneus
2. it is an ellipsoid type of synovial joint
3. tibialis posterior is the main plantar flexor
4. the main dorsiflexors are tibialis anterior and extensors of the toes

117. The deltoid ligament

1. is attached to the lateral malleolus
2. forms part of the articular surface for ankle joint

Ans. 111. B 112. C 113. A 114. D 115. E 116. D

3. is crossed superficially by tibialis anterior
4. is so strong a ligament that sprains of the ankle joint are very rare

118. The spring ligament

1. connects the dorsal aspects of the calcaneus and the navicular bones
2. forms part of the articular surface for head of talus in talocalcaneonavicular joint
3. supports transverse arch of foot
4. is supported by the tendon of tibialis posterior

119. The blood supply to the head of femur, before fusion of its epiphysis with the diaphysis, comes from

1. lateral and medial circumflex femoral arteries
2. acetabular branch of obturator artery
3. inferior gluteal artery
4. acetabular branch of medial circumflex femoral artery

120. The quadriceps femoris is

1. supplied by the femoral nerve
2. inserted into tibial tuberosity only
3. the main extensor on the knee joint
4. kept contracted in erect posture

121. The sartorius

1. takes origin from the anterior inferior iliac spine
2. lateral border forms the lateral boundary of femoral triangle
3. is inserted into the medial border of tibia
4. flexes the thigh as well as the leg

122. The adductor canal

1. lies in the middle third of the thigh
2. is bounded anteromedially by adductor longus and adductor magnus
3. transmits femoral vessels
4. transmits great saphenous vein and saphenous nerve

123. The adductor longus muscle

1. has a double nerve supply
2. is a hamstring muscle

3. is inserted mainly into adductor tubercle
4. separates femoral vessels from profunda femoris vessels

124. The gluteus maximus

1. is supplied by the inferior gluteal nerve
2. is the main extensor on the hip joint
3. is also an extensor on the knee joint
4. is partly inserted into the gluteal tuberosity

125. The tibialis anterior

1. is inserted into all the tarsal bones except talus
2. is invertor of foot
3. is supplied by the superficial peroneal nerve
4. is an important muscle for maintenance of the medial longitudinal arch

126. The popliteus muscle

1. has intracapsular origin from the medial surface of lateral condyle of femur
2. is supplied by tibial nerve
3. unlocks the knee at the end of flexion movement
4. helps in preventing injury to the lateral meniscus

127. The flexor digitorum accessorius

1. arises from the calcaneus by two heads
2. is supplied by the medial plantar nerve
3. straightens the pull of tendon of flexor digitorum longus
4. lies directly under the plantar aponeurosis

128. Trendelenberg test is positive in

1. paralysis of gluteus maximus
2. paralysis of gluteus medius
3. paralysis of psoas major
4. dislocation of the head of femur

129. The second lumbrical muscle

1. is unipennate
2. crosses the lateral side of metatarsophalangeal joint of middle toe
3. is supplied by the digital branch of medial plantar nerve
4. is supplied by the lateral plantar nerve

130. The interossei muscles of the foot

1. consist of three dorsal interossei
2. consist of four plantar interossei
3. are all supplied by the deep branch of the lateral plantar nerve
4. are so arranged that two dorsal interossei are attached to the second toe

131. The plantar aponeurosis

1. resembles exactly the palmar aponeurosis
2. is attached proximally to the calcaneus
3. is a degenerated tendon of popliteus
4. helps in maintaining longitudinal arches

132. The first layer of muscles in the sole includes

1. adductor hallucis
2. abductor digiti minimi
3. flexor digitorum accessorius
4. flexor digitorum brevis

133. The femoral sheath

1. is formed by fascia iliaca anteriorly
2. extends for about 4 cm in the thigh
3. contains superficial inguinal lymph nodes
4. has a medial compartment which contains connective tissue and a lymph node

134. In relation to the femoral hernia

1. it is more common in males
2. the neck of the hernial sac lies medial to the pubic tubercle
3. coverings from within outward are femoral septum, cribriform fascia and femoral sheath
4. the femoral ring should not be enlarged laterally to relieve strangulation

135. The femoral artery

1. is not covered by any muscles in the femoral triangle
2. continues as popliteal artery
3. pierces the adductor magnus muscle at the junction of middle and lower thirds of the thigh
4. can be felt pulsating at the mid-inguinal point

Ans. 130. D 131. C 132. C 133. C 134. D 135. E

- 136. The popliteal artery**
1. terminates at the upper border of the popliteus muscle
 2. runs down across the anterior surface of the popliteus muscle
 3. is crossed from medial to lateral side by the popliteal vein and the common peroneal nerve
 4. gives off several branches which participate in geniculate anastomosis
- 137. The lateral plantar artery**
1. is a branch of the peroneal artery
 2. enters the sole under cover of the abductor hallucis
 3. runs deep to the flexor digitorum accessorius
 4. forms plantar arch
- 138. The superficial inguinal lymphnodes drain lymph from the**
1. lower half of the anterior abdominal wall
 2. lower part of vagina and vulva
 3. lower part of the anal canal
 4. glans penis and scrotum
- 139. The lateral cutaneous nerve of thigh**
1. is a branch of sacral plexus
 2. enters the thigh deep to the inguinal ligament
 3. supplies the gluteal region
 4. is related to a condition called meralgia paraesthetica
- 140. Injury to the sciatic nerve in the gluteal region would involve**
1. hamstring muscles
 2. all the muscles of the leg
 3. all the muscles of the foot
 4. cutaneous supply along the lateral side of leg
- 141. Section of the tibial nerve may result in sensory loss along the**
1. lateral side of leg
 2. plantar surface of heel
 3. medial border of foot on the dorsal aspect
 4. entire sole
- 142. The tibial nerve**
1. runs obliquely downwards and medially across the popliteal fossa

Ans. 136. D 137. C 138. A 139. C 140. E 141. C

2. crosses deep to the popliteal artery
 3. supplies all the muscles of the posterior compartment of leg except plantaris
 4. gives branches to the knee joint
- 143. The common peroneal nerve**
1. has a root value of L_4-S_2
 2. runs downwards and laterally along the lateral border of the biceps femoris muscle
 3. supplies branches to the knee joint
 4. injury would not result in any sensory loss
- 144. The lateral plantar nerve (or its branches)**
1. supplies greater part of the skin of the sole
 2. supplies most of the muscles of the sole
 3. supplies all the nail beds
 4. is accompanied by the lateral plantar artery which lies lateral to it
- 145. Loss of sensation along on the lateral border of sole may indicate injury to**
1. first sacral nerve
 2. second sacral nerve
 3. lateral plantar nerve
 4. sural nerve

Directions: Each question consists of an assertion and a reason. Responses should be chosen as follows:

- A. if the assertion and reason are true statements and the reason is a correct explanation of the assertion;
- B. if the assertion and reason are true statements but the reason is not a correct explanation of the assertion;
- C. if the assertion is true but the reason is a false statement;
- D. if the assertion is false but the reason is a true statement;
- E. if both assertion and reason are false statements.

- 146. The range of flexion of thigh on the hip joint decreases if the knee joint is kept extended**

Because

the hamstring muscles cannot be stretched beyond a certain limit

Ans. 142. D 143. B 144. C 145. B 146. A

147. Diseases of hip joint may result in pain in the knee
Because
there are some muscles which act on both the joints
148. Fracture at the upper end of femur is always intracapsular
Because
the head of the femur is intracapsular
149. In congenital dislocation of hip joint, the displacement usually takes places on to the gluteal surface
Because
there is an opening in the posterior part of the capsule of the hip joint
150. There is a tendency for patella to dislocate medially
Because
the anterior projection of the lateral condyle of femur is more than that of the medial condyle
151. Penetrating injury at the level of the apex of the femoral triangle may result in paralysis of the quadriceps femoris muscle
Because
the femoral nerve, which is very superficial, is likely to be injured.
152. Ligation of the femoral artery above the origin of profunda femoris artery would result in ischaemia of the lower limb
Because
femoral artery is the main artery to supply the lower limb
153. The adductor magnus has a double nerve supply
Because
it is a massive muscle
154. Injury to one inferior gluteal nerve results in lurching gait
Because
gluteus medius muscle is paralysed
155. In the erect posture there is minimum activity in the extensor muscles of the hip joint
Because
flexion on the hip is prevented by the strong iliofemoral ligament

Ans. 147.B 148.D 149.C 150.D 151.E 152.D 153.B 154.E 155.C

156. **The long head of biceps femoris and the semitendinosus are flexor on the knee joint**
Because
they have a common origin from the ischial tuberosity
157. **The medial meniscus is less prone to injury**
Because
it is attached to the tibial collateral ligament
158. **Rotation on the knee joint is not at all possible**
Because
Strong collateral ligaments are present on either side of the joint
159. **Injury to the common peroneal nerve is common**
Because
it winds round the neck of fibula
160. **Injury to the tibial nerve leads to foot drop**
Because
its plantar branches supply all the muscles of sole
161. **Injury to the common peroneal nerve results in loss of dorsi flexion and eversion movements**
Because
it supplies skin on the lateral aspect of leg and dorsum of the foot
162. **Ankle joint is less stable in complete dorsiflexion**
Because
the collateral ligaments are relaxed in dorsiflexion
163. **The human foot shows longitudinal and transverse arches**
Because
arched nature of the foot provides more resilience to the foot
164. **Rupture of tendo-Achilles results in a complete loss of plantar flexion**
Because
gastrocnemius and soleus are the only plantar flexors

Ans. 156.B 157.D 158.D 159.A 160.D 161.B 162.E 163.A 164.E

165. **The range of inversion and eversion is more in plantar flexion**
Because
the narrower posterior part of the trochlear surface articulates with the tibia in this position allowing slight mobility of talus
166. **The perforating veins carry blood from deep to superficial veins**
Because
contraction of calf muscles pushes the blood into these veins
167. **Venesection and cannulation of the great saphenous vein sometimes leads to tingling sensations along the medial border of foot**
Because
the saphenous nerve is accidentally include in the ligature
168. **Fracture of the neck of femur is likely to result in avascular necrosis of the head of the femur**
Because
the head of the femur is supplied only by the retinacular arteries
169. **Intramuscular injection into the inferomedial quadrant of the gluteal region may result in foot drop**
Because
it may injure the sciatic nerve
170. **Long periods of standing predisposes to varicosities of superficial veins of lower limb**
Because
venous stasis leads to dilation of veins which reduces competency of valves